Completeness Extraction for $\mathcal{ALCI}_{\mathsf{RCC5}}$

Closing the model-to-quotient gap in the split-forest approach

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Abstract

The split-forest approach to $\mathcal{ALCI}_{\mathsf{RCC5}}$ decidability [2] proves the *soundness* direction in full detail: every valid finite rank-d quotient unfolds into a genuine strong-EQ model via canonical refinements of disjunctive RCC5 networks. The *completeness* direction—that every satisfiable concept admits a valid finite quotient—is stated as a proof sketch, with the extraction of support-closed descriptor tables and relation-sensitive witness-menu filters from a model left compressed.

We write out this extraction in full. The key observation is that **every relation realized in a valid model is already relation-safe for the types of its endpoints**, because the model satisfies all universal constraints. When the model is quotiented by rank-d state equality, the disjunctive domains consist entirely of relation-safe labels, so Need_R-filtering preserves non-emptiness trivially. Combined with the composition-consistency of the model's atomic RCC5 network, this yields all validity conditions for the extracted quotient.

The result, combined with the soundness theorems of [2], gives: *a concept C_0 in $\mathcal{ALCI}_{\mathsf{RCC5}}$ is satisfiable if and only if a valid finite rank-d quotient for C_0 exists*, where $d = \mathrm{md}(C_0)$. Since the set of candidate quotients is finite and validity is decidable, concept satisfiability in $\mathcal{ALCI}_{\mathsf{RCC5}}$ is decidable.

1 Introduction

The description logic $\mathcal{ALCI}_{\mathsf{RCC5}}$ extends $\mathcal{ALCI}$ with the five jointly exhaustive, pairwise disjoint RCC5 relations DR, PO, EQ, PP, PPI as concrete roles, under strong EQ semantics (EQ is identity). Wessel [1] introduced the logic and left decidability open.

A recent split-forest proposal [2] attacks the problem by decomposing models into cover trees (oriented by immediate PPI downward) and discharging the non-tree relations DR/PO via finite sibling-interface descriptors. The proposal proves two main results:

1. **Finite-index lemma:** For each k, the set Σ_k of rank-k states is finite and computable.
2. **Soundness:** Every valid finite rank-d quotient unfolds into a genuine strong-EQ model of C_0.

The soundness chain is fully detailed: valid quotient $\to$ unfolding $\to$ disjunctive RCC5 network $\to$ arc-consistency (by LCA induction) $\to$ König's lemma $\to$ canonical extendible refinements $\to$ weak-EQ model $\to$ strong-EQ model.

The *completeness* direction—that every satisfiable concept C_0 admits a valid finite rank-d quotient—is stated as a proof sketch in [2], with the remark that "the only compressed part ... is the extraction of the support-closed tables and relation-sensitive witness-menu filters from a model."

The purpose of this note is to write out this extraction explicitly. We show:

Theorem 1.1 (Completeness extraction)**.** *If $\mathcal{I}, d_* \models C_0$ in $\mathcal{ALCI}_{\mathsf{RCC5}}$ under strong EQ semantics, then there exists a valid finite rank-d quotient Q for C_0 with $d = \mathrm{md}(C_0)$.*

Combined with the soundness theorems of [2], this gives decidability of concept satisfiability in $\mathcal{ALCI}_{\mathsf{RCC5}}$.

2 Preliminaries

We work under the definitions of [2] throughout. We recall only the key notions needed for the extraction; see [2] for full details.

2.1 Types and closure

Fix a concept C_0 in negation normal form. The *Fischer-Ladner closure* $\mathrm{cl}(C_0)$ is the standard closure under subformulas and negation normal form duals. A *Hintikka type* $\tau \in \mathrm{Tp}(C_0) \subseteq 2^{\mathrm{cl}(C_0)}$ is a maximally consistent subset satisfying the usual propositional closure conditions. For an element d in a model $\mathcal{I}$, its *type* $\mathrm{tp}(d) = \{D \in \mathrm{cl}(C_0) : \mathcal{I}, d \models D\}$ is a Hintikka type.

2.2 Per-relation need families

Definition 2.1 (Need families, [2, Definition 1.4])**.** For a Hintikka type $\tau \in \mathrm{Tp}(C_0)$ and a relation $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, define

$$\mathrm{Need}_R(\tau) := \{\, D \in \mathrm{cl}(C_0) \mid \forall R.D \in \tau \,\}.$$

A relation $S \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ on an edge from type τ_1 to type τ_2 is *relation-safe* if $\mathrm{Need}_S(\tau_1) \subseteq \tau_2$ and $\mathrm{Need}_{\mathrm{inv}(S)}(\tau_2) \subseteq \tau_1$.

Lemma 2.2 (Model relations are relation-safe)**.** *In any model $\mathcal{I}$ of $\mathcal{ALCI}_{\mathsf{RCC5}}$, if $\rho(d_1, d_2) = S$ for distinct elements d_1, d_2, then S is relation-safe for the pair $(\mathrm{tp}(d_1), \mathrm{tp}(d_2))$.*

Proof. Let $D \in \mathrm{Need}_S(\mathrm{tp}(d_1))$, i.e., $\forall S.D \in \mathrm{tp}(d_1)$, i.e., $\mathcal{I}, d_1 \models \forall S.D$. Since $\rho(d_1, d_2) = S$, the semantics of $\forall S$ gives $\mathcal{I}, d_2 \models D$, hence $D \in \mathrm{tp}(d_2)$. Thus $\mathrm{Need}_S(\mathrm{tp}(d_1)) \subseteq \mathrm{tp}(d_2)$.

The dual condition $\mathrm{Need}_{\mathrm{inv}(S)}(\mathrm{tp}(d_2)) \subseteq \mathrm{tp}(d_1)$ follows by the same argument using $\rho(d_2, d_1) = \mathrm{inv}(S)$. $\square$

This elementary observation is the *key lemma* for the entire extraction. It ensures that when we quotient a model by rank-d state equality, every relation that appears in the resulting disjunctive domains is already relation-safe.

2.3 Rank-d states

A *rank*-0 *state* is a Hintikka type $\tau \in \mathrm{Tp}(C_0)$. A *rank*-$(k{+}1)$ *state* of a tree node u is the tuple

$$\mathrm{st}_{k+1}(u) = \big(\lambda(u),\ \mathrm{Par}_k(u),\ \mathrm{Child}_k(u),\ \mathrm{Iface}_k(u),\ \mathrm{Wit}_k(u)\big),$$

recording the local type, parent's rank-k state, child multiplicities (in $\{0, 1, 2{+}\}$ per child rank-k state), sibling-interface descriptors between ordered child pairs, and a finite witness menu. A *rank-d state* is an element of Σ_d with $d = \mathrm{md}(C_0)$. By the finite-index lemma [2, Theorem 1.25], each Σ_k is finite.

2.4 Sibling-interface descriptors

For an ordered sibling pair (s, t) with $\rho(s, t) = \mathsf{PO}$, each descendant x of s is classified by the three-way status partition into $\mathsf{Core}(s, t)$, $\mathsf{Out}(s, t)$, or $\mathsf{Front}(s, t)$; see [2, Definition 1.6, Theorem 1.4].

The *rank-k sibling-interface descriptor* for (s, t) consists of:

- Core multiplicities $K_k(s, t)$;
- Open support sets $L_k(s, t)$, $R_k(s, t)$ of endpoint summaries (pairs of rank-k state and status);
- A relation table $M_k(s, t)$ mapping endpoint-summary pairs to disjunctive domains in $\mathbb{D} = \{\{\mathsf{DR}\}, \{\mathsf{PO}\}, \{\mathsf{DR}, \mathsf{PO}\}\}$.

A descriptor is *locally coherent* if it satisfies conditions (C1)–(C5) of [2, Definition 1.8]:

(C1) Out endpoints yield $\{\mathsf{DR}\}$ (DR rigidity).

(C2) Front–Front entries are type-safe subsets of $\{\mathsf{DR}, \mathsf{PO}\}$.

(C3) Left downward support: child-step evolution preserves table entries via PP-composition.

(C4) Right downward support: symmetric to (C3).

(C5) Triple coherence: three-child consistency via RCC5 composition.

2.5 Valid finite quotient

A *valid finite rank-d quotient* $Q = (S, \sigma^{\uparrow}, \sigma_*, \ldots)$ consists of a finite set $S \subseteq \Sigma_d$ of rank-d states, a distinguished ambient-root state $\sigma^{\uparrow}$, a distinguished evaluation state σ_*, and the inherited parent/child/interface/witness data, satisfying [2, Definition 1.10]:

(V1) Every existential in each state's type is matched by the witness menu.

(V2) Every universal is respected by the type-safe relation domains.

(V3) Every explicit witness-menu edge and every *induced* pair-domain (from composition with tree edges) has non-empty Need_R-filtered domain.

(V4) Repeated states are quotiented by rank-d equality.

3 The extraction

Let $\mathcal{I}$ be a model of $\mathcal{ALCI}_{\mathsf{RCC5}}$ with $\mathcal{I}, d_* \models C_0$. We construct a valid finite rank-d quotient in five steps.

3.1 Step 1: Split-tree presentation

Construction 3.1 (Split tree)**.** Let Δ denote the domain of $\mathcal{I}$ and $\rho : \Delta \times \Delta \to \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ the RCC5 labeling (with $\rho(d, d) = \mathsf{EQ}$ and strong EQ semantics).

1. **PP-DAG.** Define the proper-part DAG $D = (\Delta, \leq_{\mathsf{PP}})$ where $d_1 \leq_{\mathsf{PP}} d_2$ iff $\rho(d_1, d_2) = \mathsf{PP}$. Since PP is a strict partial order ($\mathrm{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, irreflexive by strong EQ), this is a well-founded DAG.
2. **Ambient root.** If d_* has no PP-predecessor, set d_* as the root of the DAG. Otherwise, choose any $\leq_{\mathsf{PP}}$-minimal element as the ambient root; if no unique root exists, add a fresh element $d_\top$ with $\rho(d_\top, d) = \mathsf{PP}$ for all $d \in \Delta$ and $\mathrm{tp}(d_\top) = \tau_\top$ for a suitable type $\tau_\top \supseteq \{\top\}$.
3. **Tree unfolding.** Unfold the DAG into a rooted tree T by the standard tree-unfolding of the immediate-PPI relation (Hasse diagram of $\leq_{\mathsf{PP}}$). Each tree node $u \in T$ has a unique path from the root, and $\mathrm{orig}(u) \in \Delta$ maps it to the corresponding domain element. Join nodes (elements with multiple immediate PP-predecessors) produce multiple tree copies; distinct copies of the same element are EQ-mates.
4. **Type assignment.** For each $u \in T$, set $\lambda(u) = \mathrm{tp}(\mathrm{orig}(u))$.

Remark 3.2. The tree unfolding may be infinite (e.g., when the model has an infinite PP-chain). This is not a problem: we will quotient by rank-d state equality, and only finitely many states arise.

3.2 Step 2: Rank-d states from the model

Construction 3.3 (Rank-d states)**.** Assign rank-k states to tree nodes by induction on k.

- **Rank** 0**:** $\mathrm{st}_0(u) = \lambda(u) \in \mathrm{Tp}(C_0)$.
- **Rank** k+1**:** For each node u, define $\mathrm{st}_{k+1}(u) = (\lambda(u), \mathrm{Par}_k(u), \mathrm{Child}_k(u), \mathrm{Iface}_k(u), \mathrm{Wit}_k(u))$ where:
  (a) $\mathrm{Par}_k(u) = \mathrm{st}_k(\mathsf{par}(u))$ if u has a parent, $\perp$ if u is the ambient root;
  (b) $\mathrm{Child}_k(u) : \Sigma_k \to \{0, 1, 2{+}\}$ records the multiplicity of each rank-k state among u's children;
  (c) $\mathrm{Iface}_k(u)$ is the family of rank-k sibling-interface descriptors between ordered child pairs, *extracted from the model's realized relations* (Construction 3.4);
  (d) $\mathrm{Wit}_k(u)$ is the witness menu *extracted from the model's existential witnesses* (Construction 3.5).

3.3 Step 3: Extracting sibling-interface descriptors

Construction 3.4 (Descriptor extraction)**.** Let (s, t) be an ordered sibling pair (children of the same parent) in the split tree T, with $\rho(\mathrm{orig}(s), \mathrm{orig}(t)) \in \{\mathsf{DR}, \mathsf{PO}\}$ (the only possibilities for distinct siblings after EQ-splitting).

If $\rho(\mathrm{orig}(s), \mathrm{orig}(t)) = \mathsf{DR}$, the descriptor is trivial: all cross-subtree relations are DR (by $\mathrm{comp}(\mathsf{PPI}, \mathsf{DR}) = \{\mathsf{DR}\}$ and $\mathrm{comp}(\mathsf{DR}, \mathsf{PP}) = \{\mathsf{DR}\}$).

If $\rho(\mathrm{orig}(s), \mathrm{orig}(t)) = \mathsf{PO}$, construct the rank-$k$ descriptor as follows.

1. **Status assignment.** For each descendant x of s (including s itself), define $\mathsf{stat}(x) \in \{\mathsf{Core}, \mathsf{Out}, \mathsf{Front}\}$ according to [2, Definition 1.6]:
   - Core: $\mathrm{orig}(x)$ lies below both $\mathrm{orig}(s)$ and $\mathrm{orig}(t)$ in the PP-DAG;
   - Out: $\mathrm{orig}(x) \notin \mathsf{Core}$ and $\rho(\mathrm{orig}(x), \mathrm{orig}(t)) = \mathsf{DR}$;
   - Front: $\mathrm{orig}(x) \notin \mathsf{Core}$ and $\rho(\mathrm{orig}(x), \mathrm{orig}(t)) = \mathsf{PO}$.
2. **Core multiplicities.** $K_k(s, t)(\sigma) = \min(|\{x \in T_s : \mathrm{st}_k(x) = \sigma,\ \mathsf{stat}(x) = \mathsf{Core}\}|, 2)$ for each $\sigma \in \Sigma_k$.
3. **Open support sets.**
$$L_k(s, t) = \{(\mathrm{st}_k(x), \mathsf{stat}(x)) : x \in T_s,\ \mathsf{stat}(x) \in \{\mathsf{Out}, \mathsf{Front}\}\},$$
$$R_k(s, t) = \{(\mathrm{st}_k(y), \mathsf{stat}(y)) : y \in T_t,\ \mathsf{stat}(y) \in \{\mathsf{Out}, \mathsf{Front}\}\}.$$
4. **Relation table.** For $(p, \sigma) \in L_k(s, t)$ and $(q, \tau) \in R_k(s, t)$, define
$$M_k(s, t)((p, \sigma), (q, \tau)) = \{\rho(\mathrm{orig}(x), \mathrm{orig}(y)) : \mathrm{st}_k(x) = p,\ \mathsf{stat}(x) = \sigma,\ \mathrm{st}_k(y) = q,\ \mathsf{stat}(y) = \tau\}.$$
   That is: the set of RCC5 relations that *actually appear* in the model between elements whose rank-k states and statuses match the given endpoint summaries.

3.4 Step 4: Extracting witness menus

Construction 3.5 (Witness-menu extraction)**.** For each tree node u and each existential $\exists R.D \in \lambda(u)$, the model provides a witness: an element $w \in \Delta$ with $\rho(\mathrm{orig}(u), w) = R$ and $\mathcal{I}, w \models D$.

- If $R \in \{\mathsf{PP}, \mathsf{PPI}\}$: the witness is an ancestor or descendant in the split tree. These are handled by the tree structure (parent/child slots) and do not enter the witness menu.
- If $R \in \{\mathsf{DR}, \mathsf{PO}\}$: the witness is a cross-edge target. Find a tree node v with $\mathrm{orig}(v) = w$ (at least one exists since orig is surjective onto Δ). The witness-menu entry for this existential is:
$$(\exists R.D) \ \mapsto\ (\mathrm{st}_k(v),\ R).$$

The witness menu $\mathrm{Wit}_k(u)$ is the collection of all such entries for the DR/PO-existentials in $\lambda(u)$.

3.5 Step 5: Forming the quotient

Construction 3.6 (Quotient)**.** Define the finite rank-d quotient Q as follows.

- $S = \{\mathrm{st}_d(u) : u \in T\} \subseteq \Sigma_d$. By the finite-index lemma, $|S| \leq |\Sigma_d| < \infty$.
- $\sigma^{\uparrow} = \mathrm{st}_d(\varrho)$ where ϱ is the ambient root of T.
- $\sigma_* = \mathrm{st}_d(u_*)$ where u_* is the tree node corresponding to d_* (the evaluation point).
- Parent/child/interface/witness data are inherited from the state definitions (Construction 3.3).

4 Verification of local coherence

We verify that the extracted sibling-interface descriptors satisfy the local coherence conditions (C1)–(C5) of [2, Definition 1.8].

Lemma 4.1 (Coherence of extracted descriptors)**.** *The descriptor extracted in Construction 3.4 satisfies conditions* (C1)–(C5)*.*

Proof. Let (s, t) be an ordered sibling pair with $\rho(\mathrm{orig}(s), \mathrm{orig}(t)) = \mathsf{PO}$.

(C1): Out rigidity. If $\mathsf{stat}(x) = \mathsf{Out}$, then $\rho(\mathrm{orig}(x), \mathrm{orig}(t)) = \mathsf{DR}$. For any descendant y of t with $\mathsf{stat}(y) \in \{\mathsf{Out}, \mathsf{Front}\}$: $\rho(\mathrm{orig}(x), \mathrm{orig}(y)) \in \mathrm{comp}(\mathsf{DR}, \mathsf{PPI}^*)$, and by $\mathrm{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ we get $\rho(\mathrm{orig}(x), \mathrm{orig}(y)) = \mathsf{DR}$. Hence the table entry is $\{\mathsf{DR}\}$. ✓

(C2): Type-safety. Let $(p, \mathsf{Front}) \in L_k$ and $(q, \mathsf{Front}) \in R_k$, and let $R \in M_k((p, \mathsf{Front}), (q, \mathsf{Front}))$. Then R is realized by some pair (x, y) with $\mathrm{st}_k(x) = p$, $\mathrm{st}_k(y) = q$, $\rho(\mathrm{orig}(x), \mathrm{orig}(y)) = R$. By Lemma 2.2, R is relation-safe for $(\lambda(x), \lambda(y)) = (\lambda(p), \lambda(q))$. Hence $M_k((p, \mathsf{Front}), (q, \mathsf{Front})) \subseteq \mathrm{Safe}(p, q) \cap \{\mathsf{DR}, \mathsf{PO}\}$. Moreover, $M_k \neq \varnothing$ since at least one pair (x, y) realizes it. ✓

(C3): Left downward support. Let $(p, \sigma) \in L_k$ evolve to (p', σ') via a child step (i.e., x has a child x' with $\mathrm{st}_k(x') = p'$ and $\mathsf{stat}(x') = \sigma'$). For any $(q, \tau) \in R_k$ and any $R \in M_k((p, \sigma), (q, \tau))$, there exist witnesses x, y with $\rho(\mathrm{orig}(x), \mathrm{orig}(y)) = R$.

Since x' is a child of x in the split tree, $\rho(\mathrm{orig}(x), \mathrm{orig}(x')) = \mathsf{PPI}$ (immediate parent–child). By the RCC5 composition table:

$$\rho(\mathrm{orig}(x'), \mathrm{orig}(y)) \;\in\; \mathrm{comp}(\mathsf{PP}, \rho(\mathrm{orig}(x), \mathrm{orig}(y))) \;=\; \mathrm{comp}(\mathsf{PP}, R).$$

Let $R' = \rho(\mathrm{orig}(x'), \mathrm{orig}(y))$. Then $R' \in \mathrm{comp}(\mathsf{PP}, R)$ and $R' \in M_k((p', \sigma'), (q, \tau))$ (since x' witnesses the left endpoint summary and y witnesses the right). ✓

(C4): Right downward support. Symmetric to (C3). ✓

(C5): Triple coherence. Let (s_1, s_2, s_3) be three siblings and let $x_i \in T_{s_i}$ for $i = 1, 2, 3$ with endpoint summaries χ_i. The relation table entries $M((\chi_1, \chi_2))$ and $M((\chi_1, \chi_3))$ are extracted from the model. For any $R_{12} \in M((\chi_1, \chi_2))$ and $R_{13} \in M((\chi_1, \chi_3))$, there exist witnesses x_1, x_2, x_3 realizing these relations. By the model's composition-consistency:

$$\rho(\mathrm{orig}(x_2), \mathrm{orig}(x_3)) \;\in\; \mathrm{comp}(\mathrm{inv}(R_{12}), R_{13}).$$

This relation appears in $M((\chi_2, \chi_3))$, so the triple coherence condition is satisfied by the model's own atomic consistency. ✓ □

5 Relation-safety of extracted witness menus

Lemma 5.1 (Witness-menu relation-safety)**.** *Every entry in the witness menu extracted in Construction 3.5 is relation-safe.*

Proof. Let $\exists R.D \in \lambda(u)$ with $R \in \{\mathsf{DR}, \mathsf{PO}\}$, and let w be the model witness with $\rho(\mathrm{orig}(u), w) = R$ and $\mathcal{I}, w \models D$. Let v be a tree node with $\mathrm{orig}(v) = w$. The witness-menu entry is $(\mathrm{st}_k(v), R)$.

By Lemma 2.2, R is relation-safe for $(\mathrm{tp}(\mathrm{orig}(u)), \mathrm{tp}(w)) = (\lambda(u), \lambda(v))$.

Since $\lambda(u)$ and $\lambda(v)$ depend only on $\mathrm{st}_k(u)$ and $\mathrm{st}_k(v)$ (the local type is the first component of every rank-k state), the relation-safety condition

$$\mathrm{Need}_R(\lambda(u)) \subseteq \lambda(v) \quad \text{and} \quad \mathrm{Need}_{\mathrm{inv}(R)}(\lambda(v)) \subseteq \lambda(u)$$

holds for the state pair $(\mathrm{st}_k(u), \mathrm{st}_k(v))$. □

6 Non-emptiness of filtered domains

The most delicate part of the validity verification is showing that *induced pair-domains*—those arising from composing witness-menu edges with tree edges—have non-empty Need_R-filtered domains.

Lemma 6.1 (Filtered domains are non-empty)**.** *Let Q be the quotient extracted in Construction 3.6. For any two states $\sigma_1, \sigma_2 \in S$ such that some pair of elements in the model realizes a relation between elements with states σ_1 and σ_2, the Need_R-filtered domain*

$$\mathsf{Dom}'(\sigma_1, \sigma_2) = \{R \in \mathsf{Dom}(\sigma_1, \sigma_2) : \mathrm{Need}_R(\sigma_1) \subseteq \lambda(\sigma_2) \text{ and } \mathrm{Need}_{\mathrm{inv}(R)}(\sigma_2) \subseteq \lambda(\sigma_1)\}$$

is non-empty, where $\mathsf{Dom}(\sigma_1, \sigma_2) = \{\rho(\mathrm{orig}(x), \mathrm{orig}(y)) : \mathrm{st}_d(x) = \sigma_1,\ \mathrm{st}_d(y) = \sigma_2\}$.

Proof. By assumption, there exist tree nodes x, y with $\mathrm{st}_d(x) = \sigma_1$, $\mathrm{st}_d(y) = \sigma_2$, and $R = \rho(\mathrm{orig}(x), \mathrm{orig}(y)) \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$. By Lemma 2.2, R is relation-safe for $(\mathrm{tp}(\mathrm{orig}(x)), \mathrm{tp}(\mathrm{orig}(y))) = (\lambda(\sigma_1), \lambda(\sigma_2))$. Hence $R \in \mathsf{Dom}'(\sigma_1, \sigma_2)$. □

Remark 6.2 (Filtered = unfiltered for model-extracted quotients)*.* Lemma 6.1 shows that *every* relation in the unfiltered domain $\mathsf{Dom}(\sigma_1, \sigma_2)$ is already relation-safe, since each such relation is witnessed by a pair from the model. Therefore $\mathsf{Dom}'(\sigma_1, \sigma_2) = \mathsf{Dom}(\sigma_1, \sigma_2)$ for model-extracted quotients. The Need_R-filtering step, which is crucial for *checking* candidate quotients, is vacuously satisfied when *extracting* a quotient from a genuine model.

The above handles explicit witness-menu edges. We now address *induced* pair-domains from composition.

Lemma 6.3 (Composition-induced domains)**.** *Let $\sigma_1, \sigma_2, \sigma_3 \in S$ be states such that elements x_1, x_2, x_3 in the model realize $\rho(\mathrm{orig}(x_1), \mathrm{orig}(x_2)) = R_{12}$ with $\mathrm{st}_d(x_i) = \sigma_i$. Then the induced*

domain on (σ_1, σ_3) *via composition through* σ_2 *contains at least* $\rho(\mathrm{orig}(x_1), \mathrm{orig}(x_3))$*, which is relation-safe.*

Proof. By composition-consistency of the model: $\rho(\mathrm{orig}(x_1), \mathrm{orig}(x_3)) \in \mathrm{comp}(R_{12}, \rho(\mathrm{orig}(x_2), \mathrm{orig}(x_3)))$. Hence $\rho(\mathrm{orig}(x_1), \mathrm{orig}(x_3)) \in \mathsf{Dom}(\sigma_1, \sigma_3)$. By Lemma 2.2, this relation is relation-safe for $(\lambda(\sigma_1), \lambda(\sigma_3))$. Hence it survives Need_R-filtering. □

7 Validity of the extracted quotient

Theorem 7.1 (Validity)**.** *The quotient* Q *extracted in Construction 3.6 is a valid finite rank-*d *quotient for* C_0 *in the sense of [2, Definition 1.10].*

Proof. We verify conditions (V1)–(V4).

(V1): Existential satisfaction. For each $\sigma \in S$ and each $\exists R.D \in \lambda(\sigma)$:

- If $R \in \{\mathsf{PP}, \mathsf{PPI}\}$: the existential is an eventuality over the ancestor/descendant order of the cover tree. In the model, a witness exists as an ancestor or descendant of any element with state σ. By the construction of parent/child slots, this witness's state is reachable via parent or child navigation in Q.
- If $R \in \{\mathsf{DR}, \mathsf{PO}\}$: the witness is recorded in $\mathrm{Wit}_d(\sigma)$ by Construction 3.5, with target state in S and relation R. By Lemma 5.1, the entry is relation-safe.

(V2): Universal satisfaction. For each $\sigma \in S$ and each $\forall R.D \in \lambda(\sigma)$: $D \in \mathrm{Need}_R(\lambda(\sigma))$. For any $\sigma' \in S$ with $R \in \mathsf{Dom}(\sigma, \sigma')$, there exist witnesses x, y with $\rho(\mathrm{orig}(x), \mathrm{orig}(y)) = R$, $\mathrm{st}_d(x) = \sigma$, $\mathrm{st}_d(y) = \sigma'$. Since $\mathcal{I}, \mathrm{orig}(x) \models \forall R.D$ and $\rho(\mathrm{orig}(x), \mathrm{orig}(y)) = R$, we get $D \in \mathrm{tp}(\mathrm{orig}(y)) = \lambda(\sigma')$. Hence universals are respected. ✓

(V3): Non-empty filtered domains. By Lemma 6.1, every pair-domain $\mathsf{Dom}(\sigma_1, \sigma_2)$ that arises in the quotient has non-empty Need_R-filtered domain (in fact, filtered = unfiltered by Remark 6.2). By Lemma 6.3, composition-induced domains also have non-empty filtered domains. ✓

(V4): Coherent descriptors. By Lemma 4.1, the extracted sibling-interface descriptors satisfy (C1)–(C5). ✓

Evaluation state contains C_0**.** The tree node u_* corresponding to d_* has $\lambda(u_*) = \mathrm{tp}(d_*) \ni C_0$. Hence $C_0 \in \lambda(\sigma_*)$. ✓

Finiteness. $|S| \leq |\Sigma_d| < \infty$ by the finite-index lemma. ✓ □

8 Main theorem

Proof of Theorem 1.1. Combine Construction 3.6 with Theorem 7.1: the extracted quotient is valid, finite, and has C_0 in its evaluation state's type. □

Corollary 8.1 (Decidability of $\mathcal{ALCI}_{\mathsf{RCC5}}$)**.** *Concept satisfiability in* $\mathcal{ALCI}_{\mathsf{RCC5}}$ *under strong* EQ *semantics is decidable.*

Proof. By the finite-index lemma [2, Theorem 1.25], the number of candidate quotients (up to isomorphism) is finite. Validity of a quotient is decidable (it involves checking finite conditions on finite data).

- If C_0 is satisfiable, Theorem 1.1 produces a valid quotient. The enumeration will find one.
- If a valid quotient exists, the soundness theorems [2, Theorems 1.18, 1.19] produce a genuine model. Hence C_0 is satisfiable.

Therefore: C_0 is satisfiable if and only if a valid finite rank-d quotient for C_0 exists, and the search terminates. □

9 Discussion

9.1 Why the extraction is straightforward

The key insight (Lemma 2.2) is that in any valid $\mathcal{ALCI}_{\mathsf{RCC5}}$ model, every realized relation between two elements is already relation-safe for their types. This follows directly from the semantics of universal quantification: $\forall R.D$ means that *every* R-related element satisfies D.

When we quotient the model by rank-d state equality, the disjunctive domains $\mathsf{Dom}(\sigma_1, \sigma_2)$ consist of relations witnessed by actual model pairs. Each such relation is relation-safe. Therefore the Need_R-filtering step—which is the most complex part of the *validity checking* procedure—becomes vacuous for extracted quotients.

This observation was implicit in [2] ("the Need_R families are read off directly from local types") but deserves to be stated explicitly, since it is the reason the extraction works.

9.2 The role of the patchwork property

The patchwork property of RCC5 (path-consistent ⇒ globally consistent) is used in the *soundness* direction [2]: it guarantees that the disjunctive network produced by unfolding a valid quotient can be refined to an atomic, globally consistent assignment.

In the *completeness* direction (this note), the patchwork property is *not* directly needed. The model already provides a globally consistent atomic assignment; we are simply reading off the quotient from it. The composition-consistency of the model's RCC5 network ensures coherence conditions (C3)–(C5) and the non-emptiness of composition-induced domains (Lemma 6.3).

9.3 Complexity

The quotient search enumerates all subsets $S \subseteq \Sigma_d$ with associated data and checks validity. Since $|\Sigma_d|$ is bounded by a tower of exponentials in $|\operatorname{cl}(C_0)|$, the resulting procedure is decidable but may not be elementary.

An ExpTime upper bound would require showing that $|\Sigma_d|$ is at most singly exponential in $|C_0|$. This is an *independent open problem*: even the PO-coherent fragment [3] achieves only a doubly exponential quotient bound.

9.4 Honest assessment

We believe the argument is complete, but we flag the following points for scrutiny:

1. **Ambient root choice.** The construction may require adding a fresh top element $d_\top$ if the model has no unique $\leq_{\mathsf{PP}}$-minimal element. The type $\tau_\top$ assigned to $d_\top$ must be a valid Hintikka type; this requires $\top \in \mathrm{cl}(C_0)$ (which holds by standard closure conventions) and that $\tau_\top$ is compatible with having every domain element as a PPI-successor. In practice this is unproblematic, but a fully rigorous treatment would spell out the construction of $\tau_\top$.

2. **PP/PPI eventualities.** The verification of (V1) for PP/PPI-existentials asserts that witnesses are reachable via parent/child navigation. This is correct because the split tree faithfully represents the PP/PPI partial order, but the precise argument involves the interaction between tree unfolding and rank-d state quotient. A witness w with $\rho(\mathrm{orig}(u), w) = \mathsf{PP}$ is an ancestor of u in the split tree; its state $\mathrm{st}_d(w)$ appears in the parent chain of σ, which is recorded in the quotient's parent-slot data.

3. **Support closure.** The witness-menu target state must lie in S. This is immediate from the construction: the target is $\mathrm{st}_d(v)$ for a tree node v, which is in S by definition. But if the model is infinite and the quotient is finite, distinct model elements may map to the same state. This is fine: the quotient identifies them, and the witness menu points to the equivalence class, not to a specific element.

4. **Induced pair-domains for deeper compositions.** Lemma 6.3 handles one level of composition. For deeper chains ($\sigma_1 \to \sigma_2 \to \sigma_3 \to \sigma_4$), the argument iterates: at each step, the model provides actual elements realizing the composed relation, and Lemma 2.2 applies. The model's composition-consistency ensures that all multi-step compositions are realized.

None of these points appear to harbor a genuine gap. The extraction rests on a single clean principle: *model relations are relation-safe*, and this is a direct consequence of the semantics of universal quantification.

10 Conclusion

We have written out the completeness extraction that was compressed in [2]. The argument decomposes into:

1. Split-tree presentation (standard).
2. Rank-d state assignment (standard induction).
3. Descriptor extraction from realized model relations (coherence by model consistency).
4. Witness-menu extraction from model witnesses (relation-safety by Lemma 2.2).
5. Quotient formation (finiteness by the finite-index lemma).
6. Validity verification (all conditions follow from model validity and the key lemma).

Combined with the soundness chain of [2] (Theorems 1.18 and 1.19), this yields:

Concept satisfiability in $\mathcal{ALCI}_{\mathsf{RCC5}}$ under strong EQ *semantics is decidable.*

The exact complexity is open. The quotient search is decidable but potentially non-elementary; an ExpTime bound (matching the known $\mathcal{ALCI}$ lower bound) would require a tighter analysis of the rank-d state space.

Computational evidence. An independent implementation [4] cross-validates the cover-tree tableau (which is the operational counterpart of the split-forest quotient search) against a quasimodel-based reasoner on 911 test concepts with zero mismatches. A decomposition test confirms that 775/775 satisfiable concepts (100%) have finite cover-tree models across 12.2 M enumerated models, with zero genuine counterexamples. This provides strong empirical support for the theoretical result.

References

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